32×32 Approximate CAM with Tunable Hamming Distance

User guide

IHP SG13CMOS5L · V_{DD} = 1.2 V · V_{IOVDD} = 3.3 V · `VEVAL_IN` = 0.5 V · nominal clock of 100 MHz

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1 What the chip does

The chip stores up to 32 words of 32 bits. When it receives a search word, it compares it with all 32 stored words at once and returns one bit per word telling whether it is similar or not: a **match** if the number of differing bits does not exceed a threshold, and a **mismatch** if it does. The threshold is not fixed: the user programs it by writing a word into a special row called the *reference*.

Capacity	32 words × 32 bits
Comparison	Hamming distance against a programmable threshold
Interface	serial: 2 data inputs and 1 output, plus 3 operation pulses
Supplies	V_{DD} = 1.2 V (core), V_{IOVDD} = 3.3 V (pads), V_{SS} = V_{IOVSS} = 0 V, `VEVAL_IN` = 0.5 V
Logic levels at the pads	CMOS, 0 to 3.3 V
Clock	100 MHz nominal and maximum (the pad ring does not accept more), rising edge
Die and pads	1.2 mm × 1.2 mm, 19 bondpads of 70 µm × 70 µm
Write / reference / search	34 / 34 / 69 cycles (340 ns / 340 ns / 690 ns at 100 MHz)

Table 1: Main characteristics.

2 Pins and pads

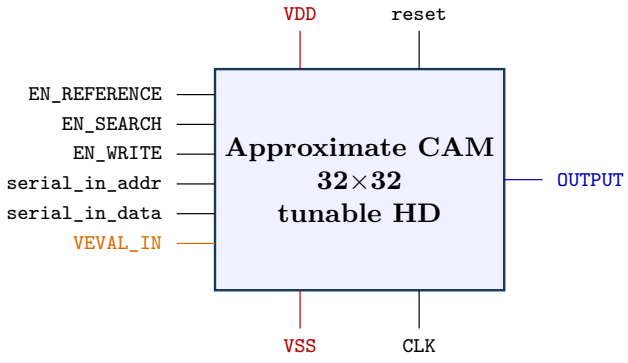


Figure 1: Chip symbol (core signals; the pad-ring supplies IOVDD and IOVSS are not drawn).

Data is always sent in its normal logical form. The chip adjusts internally the polarity the cells need, so the user does not have to invert anything.

The die has **19 bondpads** (Table 3, Figure 2): 9 signal pads and 10 supply pads (3 VDD, 2 VSS, 3 IOVDD, 2 IOVSS). Pads of the same supply are joined inside the ring, but all of them must be wire-bonded and taken to the same board net.

Pin	Type	Active	Function
VDD, VSS	Supply	–	1.2 V and ground for the logic and the CAM core.
IOVDD, IOVSS	Supply	–	3.3 V and ground of the pad ring. IOVSS is tied to VSS on the board.
`VEVAL_IN`	Analog input	–	Evaluation level of 0.5 V. Sets how strongly the match lines discharge during a search.
CLK	Input	Rising edge	System clock, 100 MHz maximum.
reset	Input	High	Returns the chip to idle and clears the registers. The pad is named `rst_n` in the layout, but it is active high.
`EN_WRITE`	Input	High pulse	Writes a word into a row.
`EN_REFERENCE`	Input	High pulse	Programs the reference, i.e. the threshold.
`EN_SEARCH`	Input	High pulse	Searches a word and delivers the result on OUTPUT.
`serial_in_addr`	Input	Serial	5-bit row address, bit 4 first.
`serial_in_data`	Input	Serial	32-bit word, bit 31 first.
OUTPUT	Output	Serial	Search result, one bit per row: 0 = match, 1 = mismatch.

Table 2: Chip pins.

#	Side	Layout name	I/O cell	Signal
1–5	West	clk, rst_n, input[0], input[1], input[2]	IOPadIn	CLK, reset, EN_WRITE, EN_REFERENCE, EN_SEARCH
6	North	analog[0]	IOPadAnalog	VEVAL_IN
7	North	input[4]	IOPadIn	serial_in_data
8–10	North	iovdd0, iovss0, iovdd2	IOPadIOVdd/IOVss	IOVDD, IOVSS, IOVDD
11	East	output[0]	IOPadOut30mA	OUTPUT
12–14	East	vdd0, vss0, vdd2	IOPadVdd/Vss	VDD, VSS, VDD
15	South	input[3]	IOPadIn	serial_in_addr
16–19	South	vdd1, vss1, iovdd1, iovss1	IOPad (supply)	VDD, VSS, IOVDD, IOVSS

Table 3: Bondpads. Numbers 1–19 are counted side by side: west bottom to top, north and south left to right, east bottom to top. Coordinates are in Table 12.

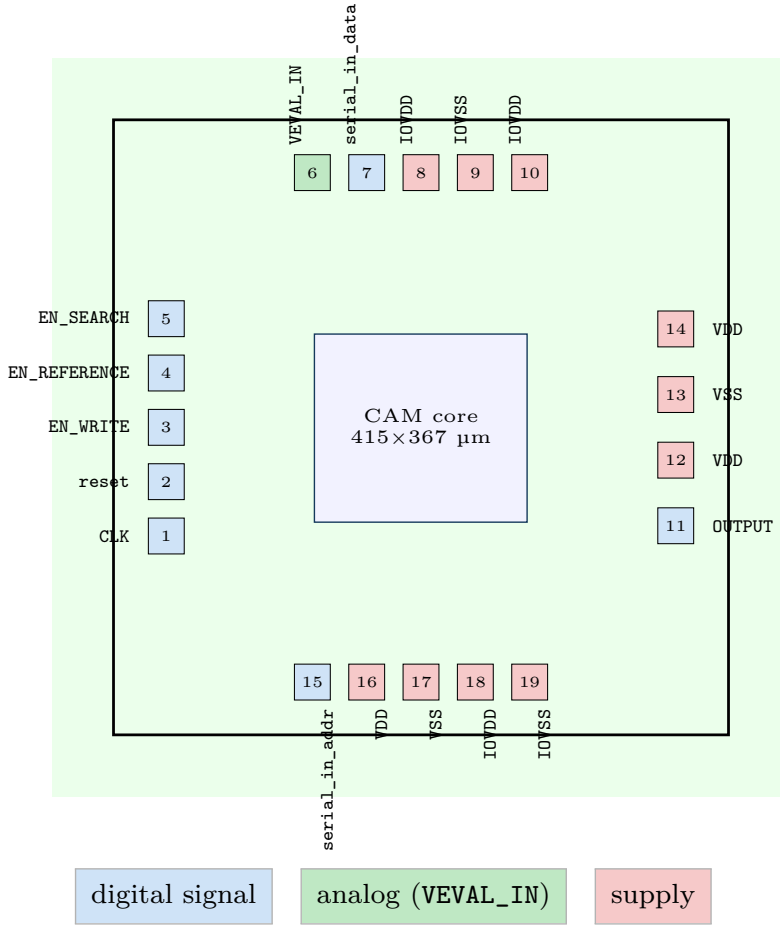


Figure 2: Pad-ring layout to scale. Numbers are those of Table 3.

3 Voltages and logic levels

- The 7 digital inputs (`reset`, the 2 serial inputs, the 3 `EN_*` and `CLK`) go through a pad supplied by IOVDD: their high level must equal IOVDD (3.3 V), not 1.2 V. The pad steps them down to 1.2 V before the core.
- If IOVDD changes, the high level of all digital stimuli changes with it: they must match.
- `VEVAL_IN` does not go through the pad level shifter: keep it fixed at 0.5 V. The analog pad adds about 587 Ω of series resistance (ESD protection), so treat it as a DC input and place a 100 nF capacitor close to its board pad.
- OUTPUT swings between 0 and IOVDD, with about 2 to 3 ns of delay through the pad. A small load to ground (a few pF) is recommended. The output pad is rated for 30 mA.
- The clock must not exceed 100 MHz: the internal phases are fixed numbers of cycles and the pad ring does not accept a faster clock.

Pin	Voltage	What it is
VDD	1.2 V DC	Core supply (CAM and digital control).
IOVDD	3.3 V DC	Pad supply. Sets the level of the inputs and of the output.
VSS, IOVSS	0 V	Common ground. All ground pads go to the same board net.
`VEVAL_IN`	0.5 V DC	Analog input. It goes straight to the core: **never more than** 1.2 V.

Table 4: Supplies and ground.

Chip pin	Signal	Type	Level
`rst_n_PAD`	reset (active high)	Digital input	0 / 3.3 V
`clk_PAD`	CLK	Digital input	0 / 3.3 V
`input_PAD[4]`	serial_in_data	Digital input	0 / 3.3 V
`input_PAD[3]`	serial_in_addr	Digital input	0 / 3.3 V
`input_PAD[2]`	EN_SEARCH	Digital input	0 / 3.3 V
`input_PAD[1]`	EN_REFERENCE	Digital input	0 / 3.3 V
`input_PAD[0]`	EN_WRITE	Digital input	0 / 3.3 V
`output_PAD`	OUTPUT (0 = match, 1 = mismatch)	Digital output	0 / IOVDD
`analog_PAD`	VEVAL_IN	Analog input	0.5 V DC

Table 5: Signal levels at each pad.

4 Test equipment

Table 6 lists what is needed to operate the chip, with example instruments. The chip needs three DC levels (1.2 V, 3.3 V and 0.5 V), a clock, a source for the five digital inputs and a way to capture OUTPUT. Two setups are proposed: a low-cost one with a USB instrument (Analog Discovery 2 or 3) for first power-up and slow functional tests, and one with an FPGA (Zybo Z7) to run at the nominal 100 MHz.

Function	Requirement	Example equipment
Supplies VDD, IOVDD	1.2 V and 3.3 V, adjustable, with current limit and readout	Two-channel bench supply (e.g. Rigol DP832, Keysight E36312A). The positive supply of the Analog Discovery (0.5–5 V) can provide IOVDD = 3.3 V.
VEVAL_IN source	0.5 V, mV resolution, low noise, current below 1 mA	SMU (Keithley 2400/2450), a third low-power channel, or a waveform-generator output of the Analog Discovery set to DC 0.5 V.
Clock	3.3 V CMOS square wave, 50% duty cycle, up to 100 MHz	Zybo Z7: PLL/MMCM from its 125 MHz reference. Analog Discovery 2/3: digital pattern output, limited to a few tens of MHz. A function generator with a 3.3 V logic output also works.
Digital inputs (EN_*, reset, data and address)	5 outputs at 3.3 V, synchronized with CLK	Zybo Z7 (Zynq-7000), high-speed Pmod ports JB/JC/JD; or the pattern generator of the Analog Discovery.
OUTPUT capture	sampling on the falling edge of CLK	Zybo Z7: FPGA logic plus Vivado ILA at 100 MHz; or the logic analyzer of the Analog Discovery for slow clocks.
Monitoring	CLK and OUTPUT integrity	Oscilloscope of 500 MHz or more; multimeter to check the supplies at the pins.
Board	socket or chip-on-board	100 nF + 10 µF decoupling on each supply group, 33 Ω series resistors on the inputs, SMA for CLK, test points.

Table 6: Equipment needed to operate the chip.

	Analog Discovery 2	Analog Discovery 3	Zybo Z7-10 / Z7-20
Digital I/O	16 channels, LVCMOS 3.3 V outputs	16 channels, 3.3 V	32 / 40 FPGA I/O on Pmod ports (3.3 V)
Pattern generator / logic analyzer	100 MS/s	125 MS/s	FPGA logic, ILA
Max. usable clock	about 25–50 MHz	about 30–60 MHz	100 MHz (nominal)
Power supplies	±0.5 to ±5 V	±0.5 to ±5 V	–
Waveform generator	2 channels, can output DC levels	2 channels, ±5 V	–
Use in this guide	power-up and slow functional tests	power-up and slow functional tests	full-speed operation

Table 7: Comparison of the suggested instruments (manufacturer data, Digilent). The clock limit of the USB instruments comes from their sampling rate: at least two samples per clock period are needed.

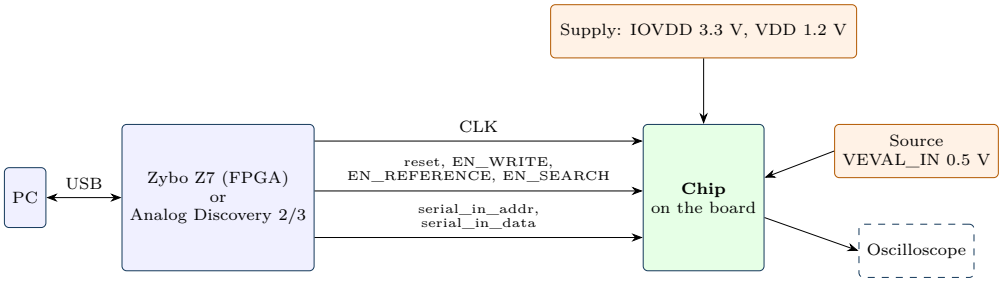


Figure 3: Test setup. All digital inputs change on the falling edge of CLK and OUTPUT is sampled on the falling edge.

5 Power-up

1. Set the current limits (VDD and IOVDD 50 mA, `VEVAL_IN` 1 mA). Connect VSS and IOVSS to a common ground. Place 10 kΩ resistors to ground on `reset` and on `EN_WRITE`, `EN_REFERENCE` and `EN_SEARCH`, so they do not float during power-up.
2. Apply IOVDD = 3.3 V first, then VDD = 1.2 V and finally `VEVAL_IN` = 0.5 V. With the supplies off, do not apply `VEVAL_IN` or any input.
3. Start CLK at 100 MHz or less, with 50% duty cycle.
4. With all `EN_*` at 0, hold `reset` at 1 for at least 2 clock cycles and then release it.
5. Program the reference (Section 7), write the words (Section 8) and search (Section 9).
6. To power down, remove `VEVAL_IN`, then VDD, then IOVDD.

After a reset or a power-up, the contents of the rows and of the reference are unknown: they must be written again.

6 Serial protocol

- The chip captures each input on the rising edge. Change the inputs on the falling edge.
- An operation starts at the first rising edge at which the chip sees its `EN_*` high while idle. That edge is called E0, and the following ones E1, E2, etc.
- An `EN_*` pulse of one or two cycles is enough. Each pulse launches a single operation; to repeat it, the pin must go low and rise again.
- Bit number k of a serial input, counting from $k = 0$, is captured at edge E$(k + 1)$.

- The next operation may start 35 cycles after a write or a reference, and 70 cycles after a search. A pulse arriving earlier is ignored.
- If several `EN_*` arrive together, the priority is: write, reference, search.

7 Programming the threshold

In a search, the reference is always compared against the fixed pattern `0xFFFF0000`. The number of bits in which the reference word R differs from that pattern is called m_{ref} and sets the threshold:

$$\text{match (OUTPUT} = 0) \iff \text{differing bits} \leq m_{ref} - 1.$$

To build R, start from `0xFFFF0000` and invert m_{ref} bits.

Recommended setting: $m_{ref} = 16$, $R =$ **`0xFFFFFFFF`.** This sets the threshold at 50% of the word (16 of 32 bits): a row that differs from the search word Q in 15 bits or fewer is a match and gives OUTPUT = 0, and one that differs in 16 bits or more is a mismatch and gives OUTPUT = 1.

Procedure:

1. Pulse `EN_REFERENCE`.
2. Send the 32 bits of R on `serial_in_data`, from bit 31 to 0, captured from E1 to E32. `serial_in_addr` is not used.
3. The reference is stored at E34 and kept for all following searches, until it is reprogrammed or a reset occurs.

Checking the threshold on the chip. On silicon the threshold may shift by one or two bits. With $m_{ref} = 16$, the OUTPUT transition from 0 to 1 must appear between the row at distance 15 and the row at distance 16 (this is what Section 10 checks). If it appears elsewhere, raise or lower m_{ref} one step at a time; each unit moves the threshold by about one bit. A higher `VEVAL_IN` also moves it, so keep `VEVAL_IN` at 0.5 V and adjust only with the reference.

m_{ref}	Word R	Match if Hamming distance	Note
15	`0xFFFF7FFF`	≤ 14	by the rule
16	`0xFFFFFFFF`	≤ 15	**recommended; verified**
17	`0xFFFEFFFF`	≤ 16	by the rule
18	`0xFFFCFFFF`	≤ 17	by the rule
19	`0xFFF8FFFF`	≤ 18	by the rule
20	`0xFFF0FFFF`	≤ 19	by the rule

Table 8: Commonly used reference words and their threshold. Only $m_{ref} = 16$ was verified in simulation of the final layout; the other values follow the rule above and should be checked on the chip.

8 Writing a word

1. Pulse `EN_WRITE`.
2. Send the 32 bits of the word on `serial_in_data`, from bit 31 to 0, captured from E1 to E32.
3. At the same time, during the first five cycles, send the row address on `serial_in_addr`, from bit 4 to 0, captured from E1 to E5.
4. The word is stored at E34.

9 Searching and reading the result

1. Pulse `EN_SEARCH`.
2. Send the search word on `serial_in_data`, from bit 31 to 0, captured from E1 to E32.
3. Wait five cycles, from E32 to E37, while the chip compares the 32 rows (pre-discharge, precharge and evaluation).
4. Read OUTPUT for 32 cycles, from E37 to E69. The bit of row i is valid between E$(68 - i)$ and E$(69 - i)$; read it on the falling edge of that cycle. Row 31 comes out first and row 0 last.
5. Interpret: **0 = match, 1 = mismatch**. Bits of rows that were not written have no meaning.

Sampling margin. OUTPUT lags the clock by the logic delay plus about 2 to 3 ns of the output pad, and the falling edge leaves only 5 ns inside the cycle at 100 MHz. On the board, add the cable or trace delay and, if the result looks shifted, move the sampling instant within the valid cycle (for example, sweep it in steps of 1 ns) until all bits are stable.

10 Functional test

Conditions: power-up per Section 5, `VEVAL_IN` = 0.5 V, room temperature, reference $m_{ref} = 16$. Repeat at 100 kHz, 1 MHz, 50 MHz and 100 MHz (the sequence was also simulated at these frequencies; USB instruments reach about 25–50 MHz, and 100 MHz requires the Zybo Z7). 100 MHz is the maximum: the pad ring does not accept more.

1. Reset for at least 2 cycles. Check that OUTPUT is 0 outside the read-out.
2. Program the reference $R =$ `0xFFFFFFFF` (Section 7).
3. Write the 32 rows. Row i ($i = 0, \ldots, 31$) stores $S_i =$ `0x0000FFFF` $\oplus (2^i - 1)$, i.e. the search word with its i least significant bits inverted. Row i is therefore at Hamming distance i from Q. For example: $S_0 =$ `0x0000FFFF`, $S_{15} =$ `0x00008000`, $S_{16} =$ `0x00000000`, $S_{31} =$ `0x7FFF0000`.
4. Search $Q =$ `0x0000FFFF` and read the 32 bits of OUTPUT.
5. **Expected result:** rows 31 to 16 give 1 and rows 15 to 0 give 0. In output order (E37 to E69) this is 16 ones followed by 16 zeros: `1111111111111111 0000000000000000`. The transition is between row 16 (mismatch) and row 15 (match).

Pass criterion: 32 of 32 bits correct at 100 MHz. If the transition appears one or two rows off, adjust m_{ref} as described in Section 7 and repeat.

Operation	Pulse at E0	`serial_in_addr`	`serial_in_data`	OUTPUT
Write	`EN_WRITE`	A4..A0, E1–E5	D31..D0, E1–E32	–
Reference	`EN_REFERENCE`	unused	R31..R0, E1–E32	–
Search	`EN_SEARCH`	unused	Q31..Q0, E1–E32	r31..r0, E37–E69

Table 9: Protocol summary, in clock edges counted from E0. The next operation may start at E35 after a write or reference and at E70 after a search.

10.1 Current measurements

Measure the VDD current in series at the VDD pad (with the supply or with a shunt resistor). IOVDD feeds the pad ring, including the 30 mA output driver: record its current separately. `VEVAL_IN` must stay below 1 mA. To measure the idle current, keep the clock running with all `EN_*` at 0, after applying and releasing reset.

As an order of magnitude, the post-layout simulation of the core without pads gives about 220 µW on average during a search at 100 MHz (about 0.2 mA at 1.2 V) and about 1.4 µW of leakage with the clock stopped. The IOVDD current was not characterized. A VDD current of several milliamperes at power-up points to a short or a wrong connection, not to normal operation.

11 Simulation and repository files

The repository already contains the testbenches; this section only says how they relate to the guide.

- The top-level netlist is `chip_top`. Its ports are those of Table 5 plus IOVDD and VSS (IOVSS and VSS are the same node in the netlist).
- `ctrl_in6_top.spice` (the three `EN_*` and `reset`) and `ser_in6_ref16_top.spice` (`serial_in_addr` and `serial_in_data`) drive the six-row test with $m_{ref} = 16$, query $Q =$ `0x0000FFFF`, at 3.3 V levels, for 0 to 3400 ns. Use them together, with IOVDD = 3.3 V, VDD = 1.2 V, `VEVAL_IN` = 0.5 V and a 100 MHz clock at 0/3.3 V whose rising edges fall at 5 ns + 10 ns $\cdot n$.
- In that test the search starts at E0 = 2625 ns, so row i is read on the falling edge at $3310 - 10\,i$ ns.

Row	Differing bits	Read at (ns)	Expected OUTPUT
31	32	3000	1
22	18	3090	1
20	16	3110	1
13	15	3180	0
5	13	3260	0
0	0	3310	0

Table 10: Expected result of the six-row test (match = 0, mismatch = 1). Rows that were not written give no meaningful value.

This six-row test was run on the final chip netlist. The full 32-row test of Section 10 was verified on the core.

12 Troubleshooting

Symptom	What to check first
OUTPUT never changes (all 0 or all 1)	`VEVAL_IN` measured at the board pad (0.5 V); `reset` released (it is active high); CLK reaches the chip at 0/3.3 V; the reference was programmed after the last reset.
No operation seems to start	`EN_*` high level equal to IOVDD (3.3 V, not 1.2 V); `EN_*` changes on the falling edge and lasts one or two cycles; no earlier operation still running (35 cycles after write or reference, 70 after search).
Result shifted by one or more bits in the sequence	Sampling instant (Section 9); row order (row 31 first); E-edge counting from E0; board and cable delay on OUTPUT.
Transition one or two rows off	Adjust m_{ref} one step at a time (Section 7); do not change `VEVAL_IN`.
High current at power-up	All 19 bondpads wired and to the right nets; IOVDD and VDD not swapped; `VEVAL_IN` not above 1.2 V; supply order of Section 5.
Wrong values only in some rows	Row was written after the last reset; address bit order (bit 4 first); rows never written give no meaning.

Table 11: Common problems and first checks.

13 Chip-on-board mounting and bonding diagram

The chip is mounted directly on the board (chip-on-board), with wires from the 19 bondpads to PCB pads. Figure 7 shows the diagram and Table 12 the coordinates of each bondpad (center, in µm, origin at the lower-left corner of the die). Pads of the same supply may be joined on the board; all VDD, VSS, IOVDD and IOVSS pads must be wire-bonded.

#	Signal	x (µm)	y (µm)	Level / board connection
1	CLK	103	388	clock, 0 / 3.3 V
2	reset	103	494	0 / 3.3 V (active high)
3	EN_WRITE	103	600	0 / 3.3 V
4	EN_REFERENCE	103	706	0 / 3.3 V
5	EN_SEARCH	103	812	0 / 3.3 V
6	VEVAL_IN	388	1097	0.5 V DC
7	serial_in_data	494	1097	0 / 3.3 V
8	IOVDD	600	1097	3.3 V
9	IOVSS	706	1097	ground
10	IOVDD	812	1097	3.3 V
11	OUTPUT	1097	408	0 / IOVDD
12	VDD	1097	536	1.2 V
13	VSS	1097	664	ground
14	VDD	1097	792	1.2 V
15	serial_in_addr	388	103	0 / 3.3 V
16	VDD	494	103	1.2 V
17	VSS	600	103	ground
18	IOVDD	706	103	3.3 V
19	IOVSS	812	103	ground

Table 12: Bondpad coordinates (center; 70 µm × 70 µm bondpad). The pad pitch is 106 µm on the north, south and west sides, and 128 µm on the east side.

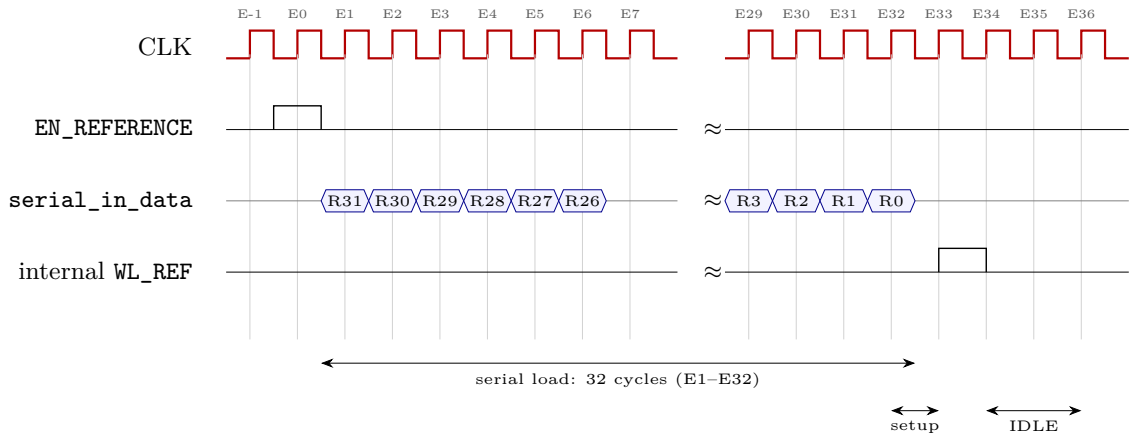


Figure 4: Reference-programming protocol. WL_REF is not a pin: it shows when the data is stored.

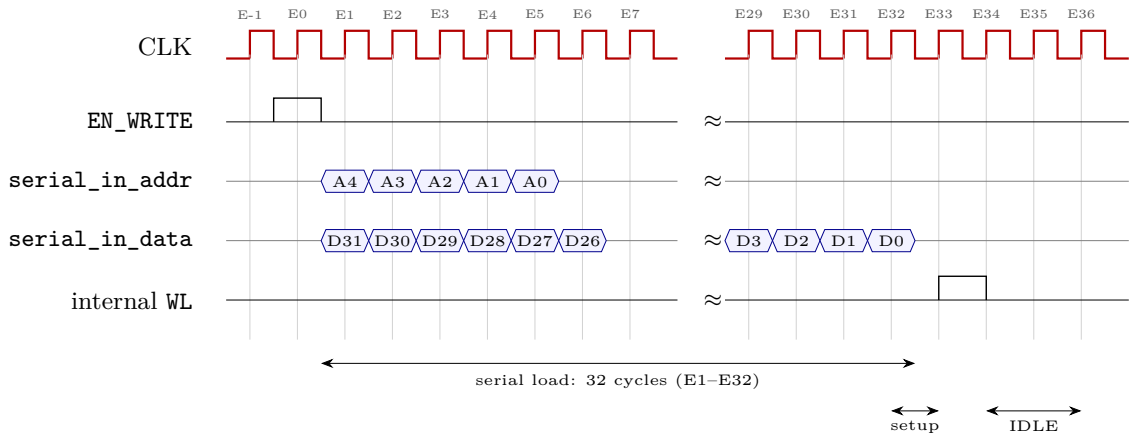


Figure 5: Write protocol. The internal WL signal is not a pin: it shows when the data is stored.

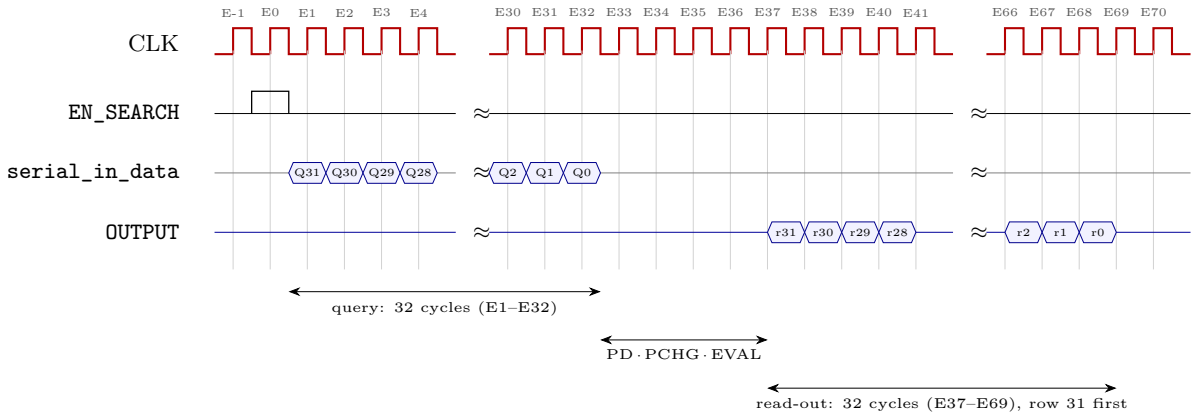


Figure 6: Search and read-out. r31 to r0 are the results of rows 31 to 0. PD = pre-discharge, PCHG = precharge, EVAL = evaluation.

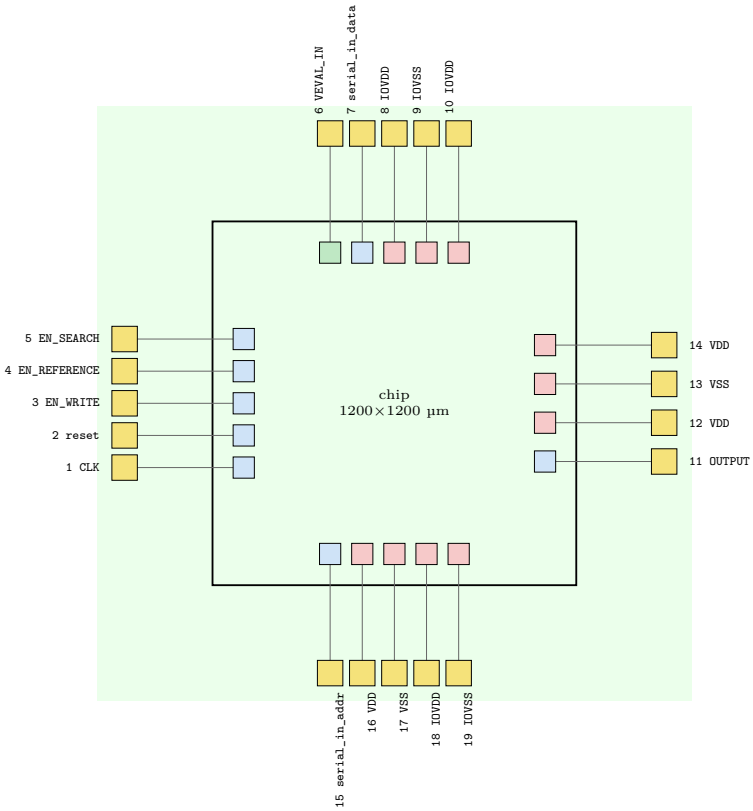


Figure 7: Chip-on-board bonding diagram. Each bondpad (colored square) is wired to a PCB pad (yellow square). The PCB pad positions are a sketch; they are defined when designing the board.